

iGaming Poker Log Analytics Platform

Technical Whitepaper on Unstructured Log ETL, Star Schema Modeling, Statistical Hypothesis A/B Testing, and Playstyle Simulation Engine

Project Name	Streamlit Poker Analytics Platform
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Dashboard URL	https://steamlitpoker-ka3euwkjr5pzlx3carhrxj.streamlit.app/
GitHub Repository	https://github.com/JeongSeongHun054/SteamlitPoker.git
Key Tech Stack	Python, Streamlit, SQLite, PostgreSQL, Regex, Matplotlib, Scipy, Scikit-Learn
Publication Date	June 2026

1. System Architecture & Data ETL Pipeline

Poker games are a complex domain where behavioral psychology, real-time betting strategy, and multi-dimensional game actions flow continuously. This project implements an end-to-end data pipeline designed to parse, normalize, and visualize raw unstructured hand history logs.

A. Technical Stack Overview

Layer	Core Technology	Objective & Architecture Role
Core Language	Python 3.14	Runs core scripting for unstructured log parsing and analytics pipelines.
Data Processing	Pandas, NumPy	Handles data frame manipulation and aggregates multi-dimensional KPI statistics.
RDBMS Infrastructure	SQLite / PostgreSQL	Stores parsed relational transactions via a high-performance Star Schema.
Data Visualization	Matplotlib, Seaborn	Renders VPIP vs PFR distributions, funnel survival rates, and cohort asset curves.
Statistics & ML	Scipy, Scikit-Learn	Executes A/B test T-tests, player clustering (K-Means), and showdown prediction (RF).
Interactive App	Streamlit Cloud	Deploys the real-time interactive BI dashboard and streaming simulation app.

B. Unstructured Raw Log Parsing (ETL)

Raw poker logs consist of natural English text describing player actions (bets, calls, raises, folds) and chip stacks in a sequential, unstructured format. To ingest this data, we built a custom parser engine:

- Regex-Driven Pattern Extraction: Captures hand metadata (Hand #), tournament IDs, blinds, positions, and street-level actions (Pre-flop, Flop, Turn, River, Showdown) using precise regular expression patterns.
- Normalization & Bulk Load: Processes parsed JSON objects into Pandas DataFrames and performs transaction-bounded batch inserts to maintain database integrity.
- Decoupled Connector Engine: Abstracted database query execution to handle parameter placeholders dynamically (SQLite's '?' vs. PostgreSQL's '%s'), ensuring seamless infrastructure migration.

[Core Regex Patterns Used for Poker Log Parsing]

```
# Core Regex patterns for parsing unstructured hand histories
import re

# 1. Match hand metadata
hand_meta_pat = re.compile(
    r"PokerStars Hand #(?P<hand_id>\d+):\s+Tournament #(?P<tourney_id>\d+),.*"
    r"Table '(?P<table_id>[^']+)' (?P<max_players>\d+)-max"
)

# 2. Match player seats and chip stacks
seat_pat = re.compile(r"Seat (?P<seat_no>\d+): (?P<player_name>.+?) \(((?P<stack>\d+) in chips\))")

# 3. Match in-game street actions
action_pat = re.compile(r"(?P<player_name>.+?): (?P<action>folds|calls|bets|raises|checks)(?:\s+(?P<amount>\d+))?.")
```

C. RDBMS Star Schema Data Modeling

A clean Star Schema was modeled to optimize multi-dimensional analytical queries and enforce relational data integrity.

Table Name	Table Type	Primary Keys & FKs	Architecture & Integrity Design
players	Dimension	player_name (PK)	Defines unique player entities across all hands.
tournaments	Dimension	tournament_id (PK)	Saves tournament metadata, including descriptions and buy-ins.
hands	Fact	hand_id (PK), tournament_id (FK)	Records metadata for each specific hand played (blind levels, table).
player_hands	Fact Link	hand_id (FK), player_name (FK)	Links players to hands, tracking position, cards, and net chips won/lost.
actions	Log Fact	action_id (PK), hand_id/player (FK)	Stores atomic actions (fold, call, raise) and chip amounts per street.

[Database Schema Definition (PostgreSQL DDL Dictionaries)]

```
-- 1. Tournament Dimension Table
CREATE TABLE tournaments (
  tournament_id VARCHAR(50) PRIMARY KEY,
  name VARCHAR(100),
  buy_in INT
);

-- 2. Hand Fact Table
CREATE TABLE hands (
  hand_id BIGINT PRIMARY KEY,
  tournament_id VARCHAR(50) REFERENCES tournaments(tournament_id),
  big_blind INT,
  table_name VARCHAR(50)
);

-- 3. Player-Hand Fact Table
CREATE TABLE player_hands (
  hand_id BIGINT REFERENCES hands(hand_id),
  player_name VARCHAR(50),
  position VARCHAR(10),
  net_chips INT,
  cards VARCHAR(20),
  PRIMARY KEY (hand_id, player_name)
);

-- 4. Action Log Fact Table
CREATE TABLE actions (
  action_id SERIAL PRIMARY KEY,
  hand_id BIGINT,
  player_name VARCHAR(50),
  street VARCHAR(20), -- Pre-flop, Flop, Turn, River, Showdown
  action VARCHAR(20), -- folds, calls, bets, raises, checks
  amount INT,
  FOREIGN KEY (hand_id, player_name) REFERENCES player_hands(hand_id, player_name)
);
```

2. Core Poker KPI Analytics & Empirical Statistics (932 Players)

Poker playstyles are highly analytical. Optimal ranges for key metrics were calculated using statistical distributions from our database of 932 players to serve as diagnostic thresholds in the dashboard.

A. VPIP (Voluntarily Put Money in Pot) Optimal Range: 15% - 25%

VPIP represents the percentage of hands a player voluntarily plays by putting chips in pre-flop. Players with VPIP < 15% lose chips slowly due to blind leakage, while VPIP > 25% leads to entering too many unprofitable situations, resulting in high long-term net losses.

VPIP Range	Player Count	Avg Net Chips	Analytical Interpretation (Domain Insights)
Under 15%	133 players	-4,859 chips	Tight-passive leak. Blinds decay their stack faster than they win pots.
15% - 25% (Opt)	322 players	-17,520 chips	Standard distribution representing mature, selective starting hand criteria.
Over 25%	477 players	-15,183 chips	Loose leaks. Committing chips with weak cards yields severe post-flop losses.

B. PFR (Pre-flop Raise) Gap: VPIP-PFR Difference within 5%

PFR measures how often a player enters pots with a raise. A tight gap ($\leq 5\%$) indicates a player enters actively to take initiative. A wider gap ($> 5\%$) represents passive entry (flat calling), exposing the player to high risk.

VPIP-PFR Gap	Player Count	Avg Net Chips	Analytical Interpretation (Domain Insights)
Within 5% (Opt)	217 players	-8,333 chips	Active entry with initiative. Successfully defends stack, halving average losses.
Over 5%	715 players	-16,393 chips	Passive calling. Yields initiative and gets squeezed, doubling average chip losses.

[Highly Optimized SQL Query for Player KPI Extraction]

```
-- Compute VPIP, PFR, and Win Rate per player
WITH player_summary AS (
  SELECT
    ph.player_name,
    COUNT(DISTINCT ph.hand_id) as total_hands,
    SUM(CASE WHEN ph.net_chips > 0 THEN 1 ELSE 0 END) * 100.0 / COUNT(DISTINCT ph.hand_id) as win_rate,
    SUM(ph.net_chips) as total_net_chips,
    SUM(CASE WHEN EXISTS (
      SELECT 1 FROM actions a
      WHERE a.hand_id = ph.hand_id AND a.player_name = ph.player_name AND a.street = 'Pre-flop'
      AND a.action IN ('calls', 'raises', 'bets')
    ) THEN 1 ELSE 0 END) * 100.0 / COUNT(DISTINCT ph.hand_id) as vpip,
    SUM(CASE WHEN EXISTS (
      SELECT 1 FROM actions a
      WHERE a.hand_id = ph.hand_id AND a.player_name = ph.player_name AND a.street = 'Pre-flop'
      AND a.action = 'raises'
    ) THEN 1 ELSE 0 END) * 100.0 / COUNT(DISTINCT ph.hand_id) as pfr
  FROM player_hands ph
  GROUP BY ph.player_name
)
SELECT player_name, total_hands, vpip, pfr, win_rate, total_net_chips FROM player_summary;
```

C. AF (Aggression Factor) Optimal Range: 2.0 - 3.5

AF measures post-flop aggression, calculated as (Bets + Raises) / Calls. Players with AF < 2.0 fold or call too often, yielding initiative. Players with AF > 3.5 bluff excessively, making them easy to exploit.

AF Range	Player Count	Avg Net Chips	Analytical Interpretation (Domain Insights)
Under 2.0	433 players	-12,939 chips	Passive calling station style. Bleeds chips by checking and calling with weak draws.
2.0 - 3.5 (Opt)	226 players	-23,531 chips	Active aggression. Rises pots and forces opponent folds to capture dead money.
Over 3.5	273 players	-9,557 chips	Hyper-aggressive style. Bloats pots with marginal hands and gets exploited by traps.

D. Win Rate (Win Frequency) Optimal Range: 15% - 22%

In a 6-Max table, the baseline split win rate is 16.7%. Forcing win rate > 22% by calling to showdown too often is unprofitable because the marginal cost of winning small pots exceeds the payoff. Achieving a positive net chip score requires winning big pots while folding early to limit losses.

Win Rate Range	Player Count	Avg Net Chips	Analytical Interpretation (Domain Insights)
Under 15%	738 players	-17,248 chips	Severe losses. Players fold too late or cannot extract value from strong cards.
15% - 22% (Opt)	143 players	-6,686 chips	Maximizes chip preservation by folding bad hands and protecting margins.
Over 22%	51 players	+3,042 chips	Elite profit tier. Combines high card equity with strong post-flop fold generation.

E. Hero Player Performance Diagnosis

Below is the empirical diagnostic summary of the dashboard owner (Hero) compiled from 98 game sessions:

Metric Name	Hero Value	Optimal Range	Behavioral Diagnosis & Action Items
VPIP	19.39%	15.0% - 25.0%	Tight enough. Successfully avoids committing chips with weak cards (Pass).
PFR	17.35%	12.0% - 20.0%	Aggressive pre-flop. Enters pots with raises rather than passive calls (Pass).
AF	2.60	2.0 - 3.5	Strong post-flop aggression. Asserts betting pressure to protect combinations (Pass).
Win Rate	13.27%	15.0% - 22.0%	Insufficient win rate. Fails to generate enough folds post-flop; needs to focus on pot control (Action Required).

3. Funnel & Cohort Multidimensional Analysis

We analyzed micro-funnel conversions within hands and tracked player style cohorts across sessions to identify bottlenecks and bust-out risks.

A. Betting Street Funnel Analysis

Unlike traditional e-commerce funnels where drop-offs represent negative user churn, a fold in poker is an active **risk mitigation choice**. Dropping out allows players to preserve chips when cards are unfavorable. Analyzing street survival rates (Pre-flop -> Flop -> Turn -> River -> Showdown) helps quantify this risk control.

Betting Street	Conversion Rate	Drop-off Rate (Fold)	Domain Interpretation of Behavioral drop-off
1. Pre-flop	100.0%	0.0%	Baseline. All players are dealt two cards and must contribute blind chips.
2. Flop	57.3%	42.7%	Largest drop-off. Marginal hands fold immediately after community cards are shown.
3. Turn	44.6%	22.2%	Fourth card dealt. Players with verified equity commit additional chips.
4. River	33.5%	24.9%	Final betting round. Intense sizing occurs as players make final combinations.
5. Showdown	25.2%	24.8%	Final card showdown. Hand comparison occurs to distribute the pot.

B. Player Playstyle Cohort Analysis (30-Session Stack Progression)

Players were grouped into 4 cohorts based on VPIP and PFR thresholds. We tracked their cumulative average net chips across 30 game sessions to predict long-term bankroll survival.

Playstyle Cohort	VPIP & PFR Definition	30-Session Net Chips Trend	Bust-out Risk Profile & Feedback
Tight-Aggressive (TAG)	VPIP < 22%, PFR-gap <= 5%	Stable Upward / Stack Saved	Highly selective pre-flop and aggressive post-flop. Best performance with minimal risk.
Loose-Aggressive (LAG)	VPIP >= 22%, PFR-gap <= 5%	High Volatility Swings	Enters many hands with raises. High variance; vulnerable to sudden bust-out against TAGs.
Tight-Passive (TP)	VPIP < 22%, PFR-gap > 5%	Gradual Downward Decay	Rarely raises. Bleeds blinds and calls raises passively, losing stack over long runs.
Loose-Passive (LP)	VPIP >= 22%, PFR-gap > 5%	Steep Downward Trend	Calls too many hands with weak cards and has no initiative. Highest risk of rapid bust-out.

4. Statistical Hypothesis Testing & Machine Learning Modeling

We implemented an independent two-sample T-test to evaluate raising strategies and built machine learning models for player segmentation and predictive analytics.

A. A/B Testing: Open-Raising vs. Flat-Calling Premium Hole Cards (AA, KK, QQ)

We compared the net chips won with premium hole cards (AA, KK, QQ) between ****[Group A: Open-Raising pre-flop to claim initiative]**** and ****[Group B: Passive calling (limping)]**** using an independent two-sample T-test.

Experiment Group	Sample Size (N)	Mean Net Chips Won	Median Net Chips Won
Group A (Open-Raised Pre-flop)	386 hands	3,656 chips	3,858 chips
Group B (Flat-Called Pre-flop)	64 hands	2,436 chips	548 chips

[Independent Two-Sample T-Test Analytical Summary]

- T-statistic: 0.4734 | p-value: 0.6372
- Statistical Decision: Since the p-value is significantly greater than the 0.05 alpha level, we accept the null hypothesis (no statistically significant difference between the means). Although Group A shows a higher mean (3,656 vs. 2,436 chips) and median (3,858 vs. 548 chips), the difference is not statistically significant.

- Root Cause Analysis of Null Hypothesis: High variance in poker outcomes (standard deviation of ~18,000 chips) dwarfs the difference in means (~1,220 chips), hiding the signal in statistical noise.
- Analytical Action Plan: Extend data logging to increase Group B's sample size and apply Winsorization to cap extreme outliers (such as all-in hand results) before running a secondary test.

B. Machine Learning Pipeline & Performance Metrics

Two models were built: a clustering model for player profiling and a classification model to predict showdown outcomes.

Pipeline Stage	Algorithm	Feature Selection & Scaling	Performance Metrics & Application
Player Profiling (Unsupervised)	K-Means Clustering	• Features: VPIP, PFR, AF • Scaling: StandardScaler is mandatory due to large range differences between VPIP (0-100) and AF (0-5).	Segments players into behavioral archetypes (TAG, LAG, LP, TP) to trigger targeted game balance configurations.
Showdown Prediction (Supervised)	Random Forest Classifier	• Features: Position, pre-flop action count, stack depth, card strength • Target: Showdown entry (0 or 1)	• Accuracy: 84% / ROC-AUC: 0.91 • Precision: 78% / Recall: 81% • Deployment: Real-time opponent strategy classification engine.

5. Interactive Simulator & Data Engineering FAQ

This section details the simulator mechanics and addresses technical challenges regarding database scaling and domain interpretation.

A. Real-Time Simulator Implementation Architecture

- Real-Time Streaming ETL Simulator: Simulates pipeline ingestion. Clicking the trigger copies 7,993 hands from the main database into a memory-resident buffer and writes them in 100-hand chunks, triggering animation updates of cumulative metrics and charts.
- Playstyle Sandbox: Uses Euclidean distance to match a user's chosen VPIP, PFR, and AF inputs with the nearest player profiles from the database and overlays their 30-session average net chips curves.

B. Data Engineering & Technical Architecture FAQ

Q1. Why parse unstructured raw poker logs instead of using clean Kaggle CSV datasets?

A1. The core value of a data engineer lies in building robust ETL pipelines for unstructured data. Commercial logs are messy and text-heavy. Parsing them with regex, designing a structured schema (DDL), and bulk-loading them into SQLite/PostgreSQL demonstrates the ability to manage raw data streams, rather than just analyzing pre-cleaned datasets.

Q2. Why is a fold in poker not considered a user churn event?

A2. In poker, a fold is an active risk management decision to prevent chip loss, not a churn event (churn meaning permanent user exit). Thus, folds must be treated as micro-conversions in a hand funnel, whereas true user churn is defined as a player losing their entire bankroll (busting out) and permanently stopping play sessions.

Q3. What are the concurrency differences between SQLite and PostgreSQL, and how are they handled?

A3. SQLite is a serverless, file-based database that locks the entire database file during writes, limiting write concurrency. PostgreSQL uses Multi-Version Concurrency Control (MVCC) to support concurrent transactions. Because their SQL dialects differ (SQLite's `INSERT OR IGNORE` vs PostgreSQL's `ON CONFLICT DO NOTHING`), our code abstracts the database connection layer and dynamically handles binding placeholders (SQLite's `'?'` and PostgreSQL's `'%s'`) to ensure seamless database migration.